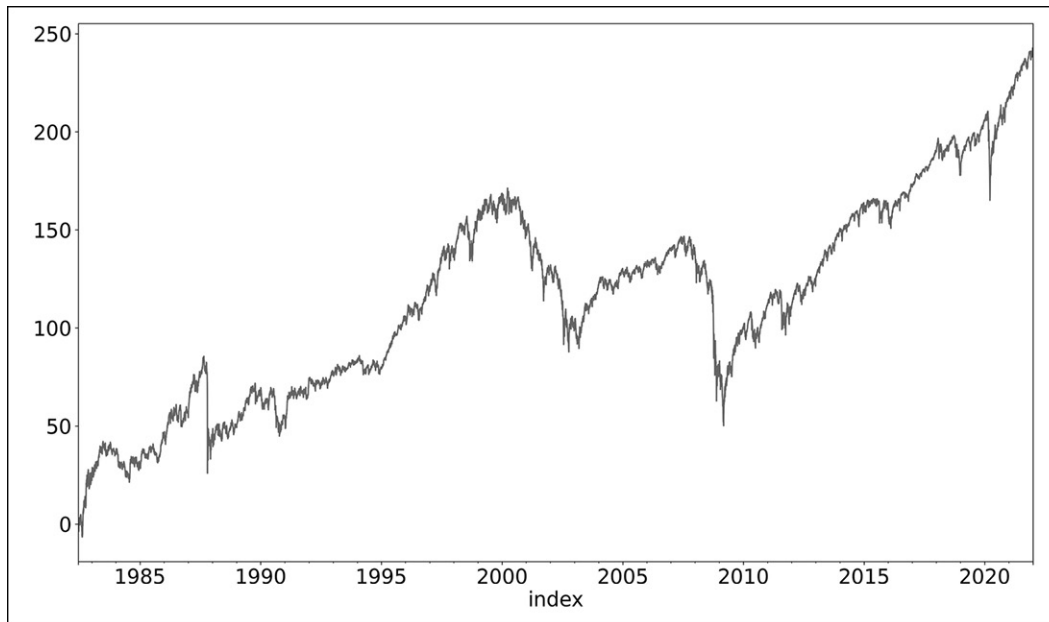




## Assessing performance characteristics

### Annual returns

Quoting performance figures as percentage returns is only the first step to properly evaluating the returns of our strategy. The total return in figure 6 is nearly 250%. That would be incredible over a single year, but we took over 40 years to earn that lucrative sum. We need to calculate an *average* annual return.

We could do this by measuring the return in each year, and taking an average. Alternatively we could measure the daily returns, take an average of those, and then *annualise* them. Assuming we're using the mean as our average, we'll get the same result.

To annualise daily returns, we need to multiply by the number of days in a year.<sup>29</sup> This is not the usual 365, the less common 366, or the approximate 365.25 (or an extremely pedantic 365.2422), since we only have returns for business days when the market is open. The number of business days per year varies depending on the year and the market you are trading but - for reasons that will become apparent below -

I prefer to assume there are exactly 256 business days in a year.<sup>30</sup>

The average daily return for the S&P 500 strategy is 0.0235%, which if I multiply by 256 equals 6.02% per year. We could also measure the average weekly, monthly or quarterly return. But I prefer to focus only on daily returns (since we're using daily data to calculate our series of profits) and annual returns (as I find them more intuitive).

## Risk and standard deviation

Futures are leveraged instruments which often have very low margin requirements. With Eurodollar interest rate futures anything up to nearly 500:1 is theoretically possible (although not sensible!). As a result we have a degree of latitude in deciding how much leverage to use. If we double our leverage, then we double our returns, but we also double our risk. So it's meaningless to talk about outright returns, rather we should discuss *risk adjusted* returns.

But it's extremely difficult to define what is meant by risk. For starters, should we use a *symmetric* or *asymmetric* measure of risk? A symmetric measure of risk treats all returns as equally risky, whether they are positive or negative. An asymmetric measure is only concerned with negative returns.

What possible logic could there be for using a symmetric measure of risk? Surely, we only worry about unexpected losses, whilst an unexpected gain is a joyous event to be celebrated? There are however a few reasons why symmetric risk makes some sense. Firstly, to get technical, if we measure our risk using all the available returns, then our risk statistic will be a more robust measure and less susceptible to occasional outliers.

Secondly, once we are using daily returns it becomes likely that we'll see both negative and positive returns: for the

S&P 500, 54% of the daily returns are positive and 46% negative. If we are currently taking a lot of risk, then we're only ever a coin flip away from a pleasingly large positive return turning into a depressingly large negative return. Hence, it makes sense to treat positive and negative returns as equally risky.

Finally, when trading futures we can be both long and short, even if for now we're limiting ourselves to long positions. If we used an asymmetric measure of risk, we'd have a different measure of price risk depending on our position. That would rather overcomplicate our lives!

With that in mind, my preferred measure for a symmetric measure of risk is the **standard deviation** of returns.

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### Why not use drawdowns?

A very popular measure of risk is the **maximum drawdown**. This is the maximum cumulative loss experienced at any point during the backtest of a trading strategy. It's an asymmetric measure of risk, which seems to make more intuitive sense. Similarly, you could use the average return divided by the maximum drawdown as a measure of risk adjusted returns. And in fact this is a very popular measure amongst many traders: it even has a name - the 'Calmar Ratio'.

But there are a few reasons why I'm not especially enamoured of the maximum drawdown as a risk measure. There is only one maximum drawdown in a data set, so our statistics are dependent on a small subset of the data.<sup>31</sup> Secondly, the size of the maximum drawdown depends on the length of your data set; the largest drawdown seen over a 30 year backtest will usually be much larger than the same statistic for a single year.

Finally, using our maximum drawdown as the risk measure

may give us a false sense of security, and fool us into thinking that this figure is the most we can lose. This is unlikely to be the case.<sup>32</sup>

We could make the measure a little more robust by using the *average* drawdown and I will also calculate this figure as it does give some insight into the returns of a strategy (to estimate this we measure the current drawdown on all the days in our data set, and then calculate the mean of those daily drawdown figures).

But for my primary risk measure I still prefer to use the standard deviation.

---

To calculate the standard deviation of a series of returns  $r_1 \dots r_T$  we'd first find the mean  $r^*$ :

$$r^* = [1 \div T]r_t + [1 \div T]r_{t-1} + [1 \div T]r_{t-2} \dots + [1 \div T]r_1$$

We can then calculate a standard deviation:

$$\sigma = \sqrt{[1 \div T](r_T - r^*)^2 + [1 \div T](r_{T-1} - r^*)^2 + [1 \div T](r_{T-2} - r^*)^2 + \dots [1 \div T](r_1 - r^*)^2}$$

We can calculate this on an annual or daily basis, or for any other frequency. We can also *annualise* the daily standard deviation of returns, as we did for average returns. This means we can use more data (over 10,000 daily returns rather than just 40 annual returns), but end up with a more intuitive annual figure.

Under certain assumptions, which I'll come to in a second, we can do this by multiplying the daily standard deviation by the square root of the number of business days in a year. If, like me, you assume there are always 256 business days in a year then we can annualise a daily standard deviation by multiplying it by the square root of  $\sqrt{256} = 16$ . This nice

round number is my reason for assuming there are precisely 256 business days in every year.

If we measure the daily standard deviation of returns in figure 5 we get 1.2%, which when annualised is 19.2% per year. But if we directly measure annual standard deviation we get 17.2%. These figures are different, and not just because the actual number of business days is not precisely 256 days in every year. They are different because the ‘certain assumptions’ that I made are not true in practice.

The first assumption I made when annualising was that there was zero *auto-correlation*: yesterday’s return has no influence on tomorrow’s return. This is unlikely to be the case in practice. In financial data we often get positive auto-correlation, which is the phenomenon by which up days tend to be followed by up days, and vice versa; or negative auto-correlation when up days are usually followed by down days, with the reverse being true.

Consider an extreme situation in which we made \$1 one day, and then lost \$1 on the subsequent day, then made \$1, lost \$1, and so on; an example of perfectly negative auto-correlation. Every year we would make \$0 or \$1, and our standard deviation measured using annual returns would be around \$1. But our daily standard deviation would be \$1, or \$16 once annualised. Negative auto-correlation of daily returns results in annualised standard deviation being too high.

The second assumption was that daily returns have a symmetric, *well-behaved* distribution. For example, suppose we earn \$1 every day, and then on the final day of the year we either lose \$364 or gain \$364. Trivially, the annual standard deviation will be \$364, since in half the years we break even and in the other half we make \$728. However the daily standard deviation is \$19.05, which when annualised comes to just \$304.

These are contrived examples designed to give extreme results, and generally the annualised figure is pretty close to the value calculated by using actual annual returns, as we can see for S&P 500. It's usually better to use the annualised figure calculated with daily returns since it uses more data points for its estimation, and thus is more robust. But you should be aware that it may sometimes give misleading results.

## Risk adjusted returns: Sharpe ratio

Now we have a measure of risk, we can use it to calculate risk adjusted returns. The **Sharpe ratio (SR)** is the average excess return (mean return less risk-free interest rate), divided by the standard deviation of returns:

$$\begin{aligned} &\textbf{Sharpe ratio (SR)} \\ &= (\textbf{Mean return} - \textbf{Risk-free rate}) \div \textbf{Standard deviation} \end{aligned}$$

In this book we exclusively use excess returns, so we do not need the risk-free rate:

$$\textbf{SR for futures} = \textbf{Mean excess return} \div \textbf{Standard deviation}$$

Note that both the mean and the standard deviation need to be estimated with the same frequency. So we can divide the average daily return by the standard deviation of returns, which for the S&P 500 strategy gives a daily SR of  $0.023\% \div 1.2\% = 0.0196$ . Or we could divide the average annual return by the standard deviation of annual returns to calculate an annual SR.

Once again, as it's better to use more data, we'd rather calculate a daily SR and annualise it. We already know we can annualise daily means by multiplying by the number of days in a year, whilst annualising standard deviations involves multiplying by the square root of the number of days. Using trivial mathematics, we can show that annualising daily SR

also requires multiplying by the square root of the number of days in a year (which I assume is 256). The annualised SR for the S&P 500 is:

$$0.0196 \times \sqrt{256} = 0.0196 \times 16 = 0.31$$

For the rest of this book, unless I specify otherwise, all Sharpe ratios I quote will be annualised.

Since we know that annualising standard deviations is fraught with potential danger, the same must be true<sup>33</sup> of annualising SR. But because standard deviations aren't usually too badly affected, the effect on SRs is also pretty minimal.

### Measuring asymmetry: Skew

Many people are critical of the assumption of symmetric returns that underpins standard deviation, which makes the Sharpe ratio guilty by association. They prefer the **Sortino ratio** which is similar, but instead of standard deviation (the average deviation of both positive and negative returns) uses the deviation of only negative returns.

This has some value when deciding which of several trading strategies to implement, but it does not help when trying to understand the behaviour of a given strategy. If a strategy or asset has a high Sortino ratio, is it because it has a high Sharpe, or because it has a low Sharpe but has fewer negative returns?

For this reason, I prefer to use the SR,<sup>34</sup> but then to measure the symmetry of returns separately. There are a few different ways to do this. The classical method for measuring symmetry is with **skew**. The returns from a *positively* skewed asset will contain more losing days than for those that are negatively skewed. But the losing days will be relatively small in magnitude. A *negatively* skewed asset will have fewer down days, but the losses on those days will be larger.

Buying insurance on your house or car is a positive skew strategy. You will experience frequent small losses (paying monthly or annual premiums) with occasional large gains (receiving payouts when you crash your car or get burgled). The insurance companies you are paying premium to will see frequent small gains on your account (receiving premiums) and occasional large losses (making payouts) - negative skew.

Right at the start of the chapter I said it was smart to take a risk that most other people are unwilling, or unable, to take. Most people strongly dislike negative skew, so taking on this risk should result in higher profits. Running an insurance firm is usually profitable, hence we'd expect negative skew to be associated with higher returns.<sup>35</sup> It's important to be aware of the degree of skew in a given asset. If an asset has strongly negative skew, then we know our assumptions about symmetrical returns will be way off the mark, and its price is likely to be highly unstable.

Skew can also be measured for different time periods. It doesn't matter so much right now, but we'll see later in the book that you can get very different skew figures when using daily, monthly or annual returns. Again, this is down to auto-correlation. For example, suppose we had a strategy with positive auto-correlation that consistently made money in the first three weeks of each month, but then lost most of it in the fourth week. The weekly returns would have serious negative skew, but the monthly returns would be just fine.

I'm mostly going to use monthly skew, since daily and weekly skew can be seriously affected by a couple of extreme daily returns, and annual skew does not give us enough data points for a reliable estimate. The skew of S&P 500 monthly percentage returns come in at  $-1.37$ . Negative skew of this magnitude is fairly typical of assets seen as risky, like equities.

## Measuring fat tails



As well as negative skew, equities also famously have *fat tails*. A fat-tailed distribution is one where extreme returns are more likely than in the standard normal Gaussian distribution. If equity returns were Gaussian, then a six standard deviation move would occur every 2.7 million years or so. With a typical daily standard deviation for S&P 500 of 1.2%, a six standard deviation move is around 7.2%. Theoretically, we would have been *very* unlucky to see a single daily return of 7.2% or more in the last 100 years. In fact there have been around 40 such days!

How can we measure these fat tails? We could continue down the route of classical statistics, by using **kurtosis**: the counterpart of skew that measures fat tails. However, I am not keen on this for a few reasons.

Firstly, I find it quite difficult to interpret figures like 6.65 (which, as you may have guessed, is the monthly kurtosis of S&P 500 futures). Also, a single figure for kurtosis does not tell you whether you have unwanted fat left tails (unusually sizeable negative returns), or extremely pleasant fat right tails (unexpectedly large positive returns). Finally, kurtosis is not a robust statistical measure, and it will swing wildly if one or two extreme days are removed or added to a data series.

Instead I'm going to use an alternative measure of fat tails. This statistic directly measures just how extreme our large returns are compared to a normal Gaussian distribution.

The first step is to *demean* the return series, by subtracting the daily mean return from every data point. Next, we measure the *percentile* of the demeaned returns at various points. Firstly, the 1st percentile: the point at which 1% of the returns in the distribution are lower, which comes in at -3.25% for the S&P 500. Then we measure the 30th percentile (-0.28%), 70th percentile (+0.42%), and 99th percentile (+3.04%).

I've chosen the 30% and 70% percentile as they're roughly equivalent to a minus one and plus one standard deviation move. The 1% and 99% points reflect the extremities of the distribution. If you wish you could use 0.1% and 99.9% to get more extreme estimates, but then you would have fewer data points contributing to your estimate, making it less reliable. This isn't so much of a problem with the 40 years of S&P 500 data, but we might only have a few years of data for some other instruments.

The next step is to divide the first two percentiles to get a lower percentile ratio:<sup>36</sup>

$$\begin{aligned}\text{Lower percentile ratio} &= \text{1st percentile} \div \text{30th percentile} \\ &= -3.25\% \div -0.28\% = 11.6\end{aligned}$$

Similarly we can calculate an upper percentile ratio:

$$\begin{aligned}\text{Upper percentile ratio} &= \text{99th percentile} \div \text{70th percentile} \\ &= 3.04\% \div 0.42\% = 7.2\end{aligned}$$

For a Gaussian normal distribution, both of these ratios will be equal to 4.43, as it's a symmetric distribution. To get an idea of how the S&P 500 compares to a normal distribution, we divide each of the ratios by 4.43 to get a *relative* ratio:

$$\begin{aligned}\text{Relative lower fat tail ratio} \\ &= \text{Lower percentile ratio} \div 4.43 = 11.6 \div 4.43 = 2.16\end{aligned}$$

$$\begin{aligned}\text{Relative upper fat tail ratio} \\ &= \text{Upper percentile ratio} \div 4.43 = 7.2 \div 4.43 = 1.60\end{aligned}$$

These are rather unwieldy phrases, so for the remainder of the book I will use these abbreviated terms: **lower tail** and **upper tail**. Each of these gives us an indication of how extreme the lowest and highest returns are. Any value higher than 1 indicates our extreme returns are more fat tailed

than a normal Gaussian distribution. To put it another way, 1% of the time with a Gaussian distribution we'd get a loss that's about 4.43 times larger than our average loss. But for the S&P 500, 1% of the time we get a loss that's more than double that, 2.16 times bigger to be precise.

## Summary statistics for S&P 500

I could continue discussing performance statistics for many more pages, and for several additional chapters, but I feel the statistics we have so far are sufficient to judge the performance and risk of any given trading strategy. Here's a summary of the statistics I've calculated for the S&P 500:

|                                         |                      |
|-----------------------------------------|----------------------|
| Strategy: Buy and hold, single contract | S&P 500 micro future |
| Years of data                           | 41                   |
| Mean annual return                      | 6.0%                 |
| Average drawdown                        | −23.2%               |
| Annualised standard deviation           | 19.2%                |
| Sharpe ratio                            | 0.31                 |
| Skew                                    | −1.37                |
| Lower tail                              | 2.16                 |
| Upper tail                              | 1.60                 |

## Buy and hold with US 10-year bond futures

We originally decided to buy and hold the S&P 500 micro future to benefit from the equity risk premium. We can earn another well-known financial risk premium from taking interest rate risk (also known as *duration* risk). We can realise this through holding bond futures, of which the US 10-year government bond is one of the most liquid.

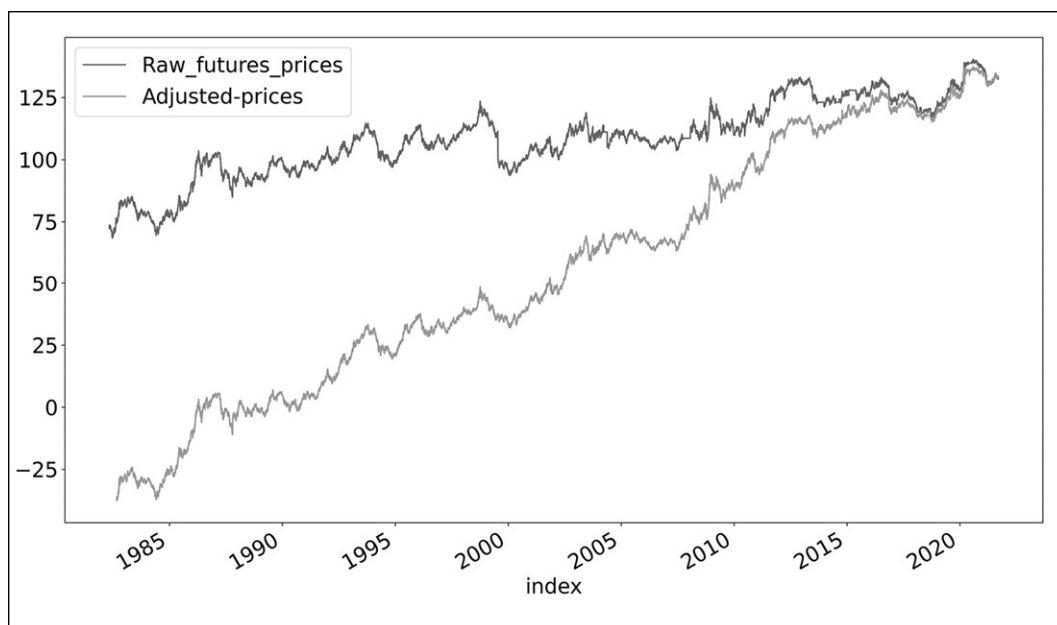
The relevant information about the contract is shown below:

|  |  |
|--|--|
|  |  |
|--|--|

|                                          |                                                           |
|------------------------------------------|-----------------------------------------------------------|
| Instrument                               | US 10-year bond future ( ‘T-note’ )                       |
| Symbol and Exchange                      | TN, CBOT                                                  |
| Futures multiplier                       | \$1,000                                                   |
| Tick size                                | 0.015625 (1/64)                                           |
| Commission                               | \$0.85                                                    |
| Current cost of trading,<br>per contract | \$8.6625                                                  |
| Delivery                                 | Physical                                                  |
| Expiry months                            | Quarterly: March, June, September,<br>December            |
| Roll pattern                             | Hold first contract until around 25<br>days before expiry |

The raw futures contract prices, and back-adjusted prices, are shown in figure 7.

Figure 7: Raw and back-adjusted price series for US 10-year bond futures



You may find this plot quite startling; the adjusted price series significantly outperforms the raw futures price, and

has done so consistently over time. Almost all of this is due to significantly higher levels of carry.

One consequence of this high carry, and the length of time we have data for, is that the back-adjusted price is actually negative for the first few years. This is nothing to worry about, as long as we ensure our strategies aren't sensitive to negative prices. In fact, the current price of crude oil futures did actually go negative in March 2020, in response to the COVID-19 pandemic. Fortunately my trading system code had been written to be robust in the presence of negative numbers and this did not cause any computer glitches!

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### Magic numbers and round numbers

An unwanted side effect of using adjusted prices is that we can't incorporate 'magic' or 'round' numbers into our strategies, because market participants wouldn't be using adjusted prices for their trading decisions but the actual futures prices that prevailed in the market at that time. Here's an example of one magic number strategy, based on the Fibonacci series:

*Suppose the price of a stock rises \$10 and then drops \$2.36. In that case, it has retraced 23.6%, which is a Fibonacci number. Fibonacci numbers are found throughout nature. Therefore, many traders believe that these numbers also have relevance in financial markets.*

Source: [investopedia.com](https://www.investopedia.com)

I am not one of these many traders. Personally, I think Fibonacci numbers are complete hokum.

Round number strategies are marginally more plausible. Most people who are putting in stops (orders to close or open positions if some price level is achieved) will